Module 04: Cognition in Robotics — State Estimation and Filtering

Learning Objectives

1. Define **state estimation** as Active Inference in the continuous domain — maintaining probabilistic beliefs about hidden states from noisy observations.
2. Analyze the relationship between **Kalman filtering, particle filtering, and variational inference** as alternative implementations of the same inference problem.
3. Apply state estimation to real robotic tasks: localization, object tracking, and sensor fusion.

Introduction

A robot's sensors are noisy, incomplete, and delayed. A LiDAR scan contains measurement error. A camera image is ambiguous about depth. An IMU drifts over time. The robot must infer the true state of the world — its own position, the positions of obstacles, the velocities of moving objects — from these imperfect observations. This is **state estimation**: the core cognitive function of any robotic Active Inference agent.

State estimation connects directly to the Free Energy Principle: the robot maintains a probabilistic belief (the approximate posterior) over hidden states and updates this belief to minimize variational free energy — the discrepancy between its model's predictions and its sensory observations.

Key Concepts

1. The Kalman Filter as Variational Inference

The **Kalman filter** — the workhorse of robotic state estimation — is a special case of variational inference under linear-Gaussian assumptions:

- **Prediction step** (B matrix): Project the current state belief forward using the transition model → produces a prior for the next timestep
- **Update step** (A matrix): Incorporate the new observation using the observation model → produces a posterior that combines prior and likelihood
- **Kalman gain** = **precision weighting**: Determines how much the observation shifts the belief, based on the relative precision of the prior vs. the observation

When the transition model is accurate (high prior precision), the Kalman gain is small — the robot trusts its predictions. When observations are highly precise (clean sensors, good conditions), the Kalman gain is large — the robot trusts its sensors. This is exactly the precision-weighted belief updating of Active Inference.

2. Extended Kalman Filter (EKF) and Nonlinear Inference

Real robotic systems are nonlinear. The Extended Kalman Filter handles nonlinearity by linearizing the transition and observation models at each timestep:

- The robot's motion model might include trigonometric functions (rotation) → nonlinear B matrix
- The camera projection model involves perspective division → nonlinear A matrix
- EKF linearizes these via Jacobians, then applies standard Kalman update

When nonlinearity is severe, the EKF approximation breaks down. The **Unscented Kalman Filter** (UKF) provides a better approximation by propagating sigma points (carefully chosen sample states) through the nonlinear functions.

3. Particle Filters for Multi-Modal Beliefs

When the posterior is multi-modal (the robot might be at location A *or* location B, but not sure which), Gaussian filters fail. **Particle filters** represent the posterior as a set of weighted samples (particles):

- Each particle is a hypothesis about the robot's state
- High-weight particles are in regions of high posterior probability
- Low-weight particles are pruned and resampled from high-weight regions
- The particle cloud naturally represents multi-modal, non-Gaussian beliefs

Monte Carlo localization (MCL) uses a particle filter for global robot localization: when the robot is first placed in a known map but doesn't know where, particles spread across the entire map. As the robot moves and observes, particles collapse onto the true location.

4. Sensor Fusion as Precision-Weighted Combination

Combining information from multiple sensors is **precision-weighted belief combination** — the Active Inference implementation of multi-sensory integration:

- Each sensor provides an observation with an associated precision (inverse variance)
- The fused belief is the precision-weighted average of individual sensor beliefs
- GPS (high precision in open sky, low precision in urban canyons) + IMU (constant moderate precision) + LiDAR (high precision in structured environments) → fused localization that inherits the best precision from each source in each condition

Applications

- **Autonomous vehicle localization**: A self-driving car fuses GPS, LiDAR, camera, radar, and wheel odometry using an EKF or particle filter to maintain a precise, real-time estimate of its position and orientation. The Kalman gain automatically shifts weight between sensors based on conditions (tunnels reduce GPS precision; fog reduces camera precision).
- **Visual-inertial odometry (VIO)**: A drone combines accelerometer/gyroscope data (high-frequency, drifting) with visual feature tracking (lower-frequency, drift-free) in a

tightly coupled filter — achieving robust state estimation for aggressive flight maneuvers.

Conclusion

State estimation is the cognitive core of robotic Active Inference — maintaining and updating probabilistic beliefs about hidden states from noisy, multi-modal observations. Kalman filters, particle filters, and sensor fusion are all implementations of the same variational inference problem. The next module examines how estimated states drive control actions.